

Introduction and Purpose

Visually impaired individuals rely on assistive technologies to interpret their surroundings. However, most existing tools are limited. These systems only focus on object recognition, lack memory, and fail to provide context or prioritize relevant information. Already existing solutions are prone to hallucinations that generate inaccurate, unreliable findings. YOLO addresses these problems by creating an AI system that helps blind people navigate their surroundings independently, and with greater confidence. It has real-time, adaptive, and intent-driven visual description, integrating scene graph generation, memory querying, and contextual reasoning. YOLO supports visually impaired users in understanding not only the moment, but also events they might miss, making everyday navigation easier.

Dataset and Experimental Setup

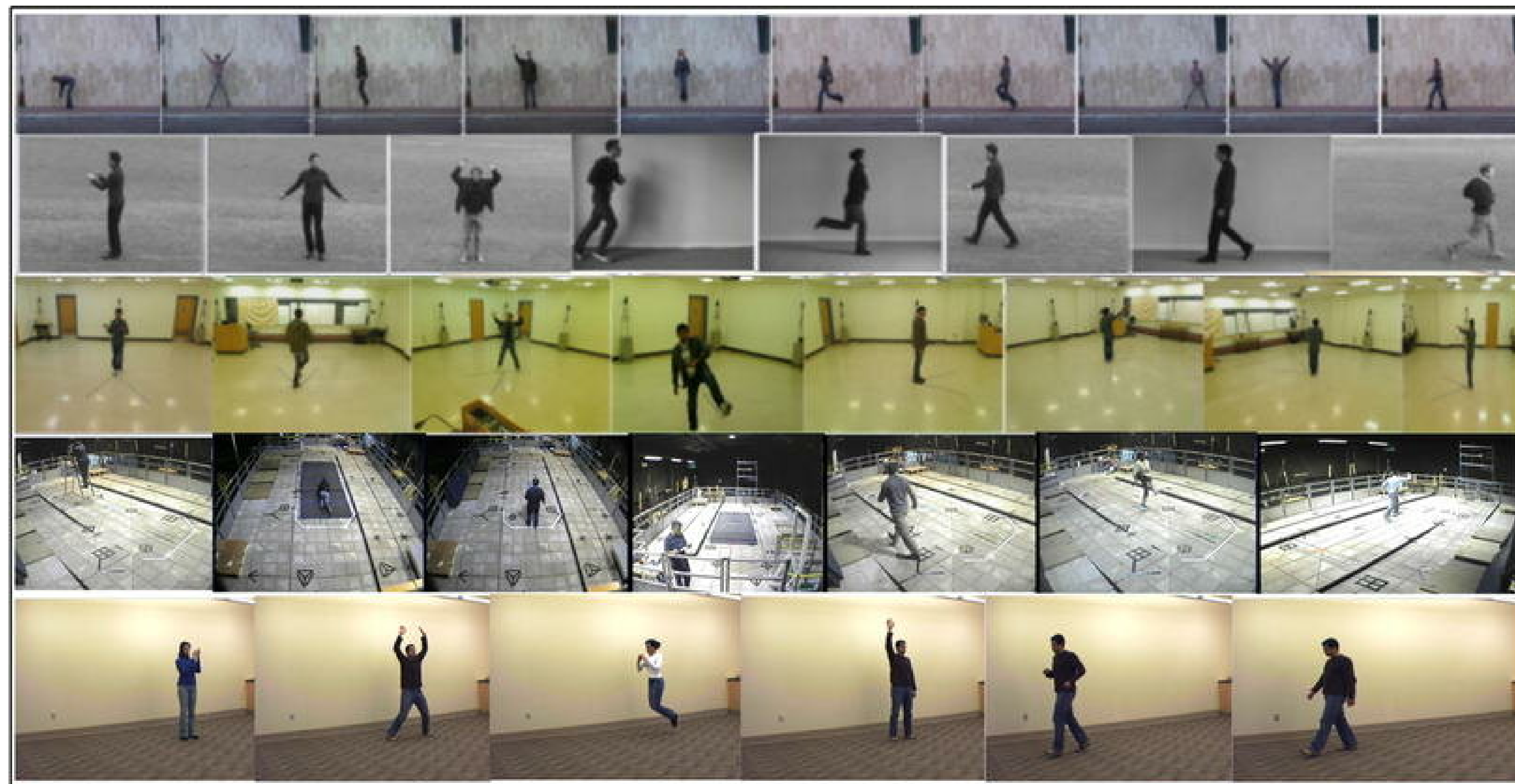


Figure 1. Sample Frames of Dataset

Datasets: STTran was trained on the Action Genome dataset, containing richly annotated video data focusing on spatial-temporal relationships between human actions and objects. G-Retriever was trained on the scene graphs of the GraphQA dataset, containing complex 3D scene-graphs detailing relationships between objects along with question-answer pairs about each of those graphs.

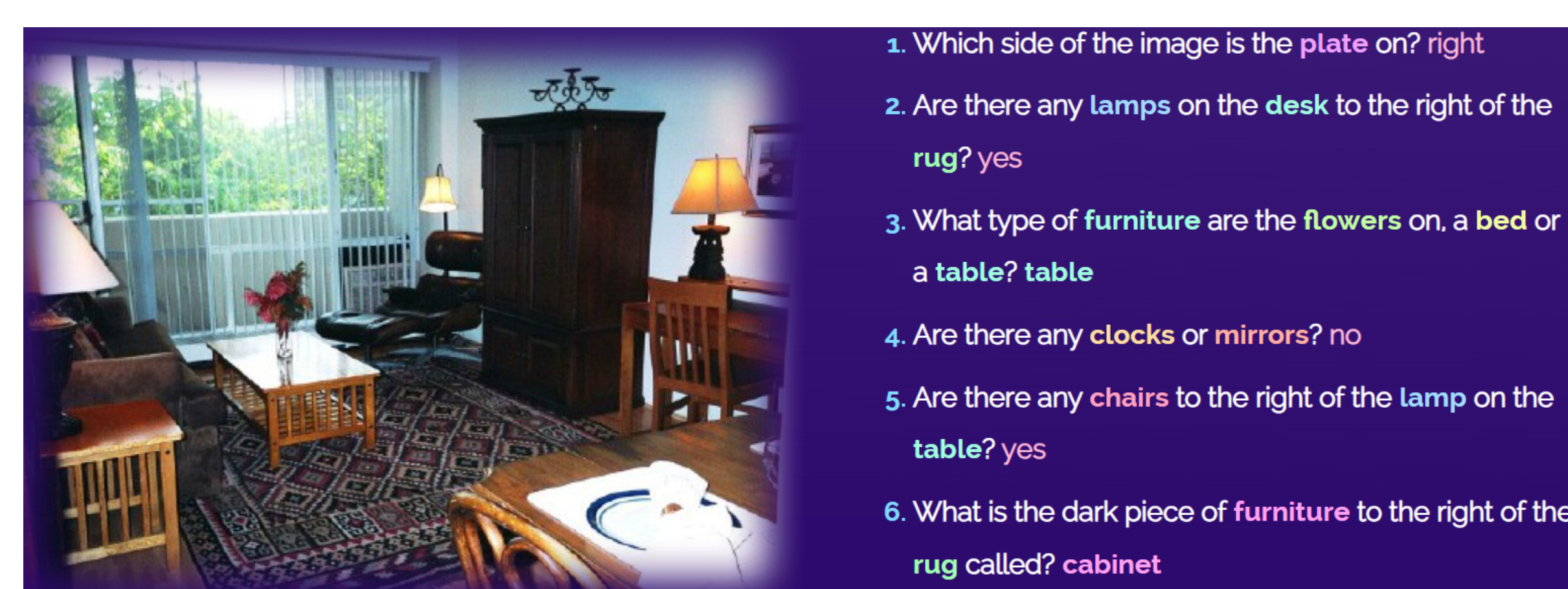


Figure 2. Sample Image with Corresponding Questions

Pipeline

STTran: Processes video frames to create structured scene graphs that represent complex environments as object-relationship-object tuples, enabling visually impaired users to receive contextually rich, query-able descriptions that maintain both spatial relationships and temporal memory.

G-Retriever: Enables graph question-answering by leveraging language capabilities of LLMs and graph-centered inductive biases of GNNs to allow users to ask natural language queries about graphs and receive relevant answers. It performs RAG to retrieve relevant nodes and edges, constructs a subgraph via a PCST algorithm, encodes the subgraph via a GNN and MLP, and uses a text-embedder and LLM to process the constructed subgraph and query.

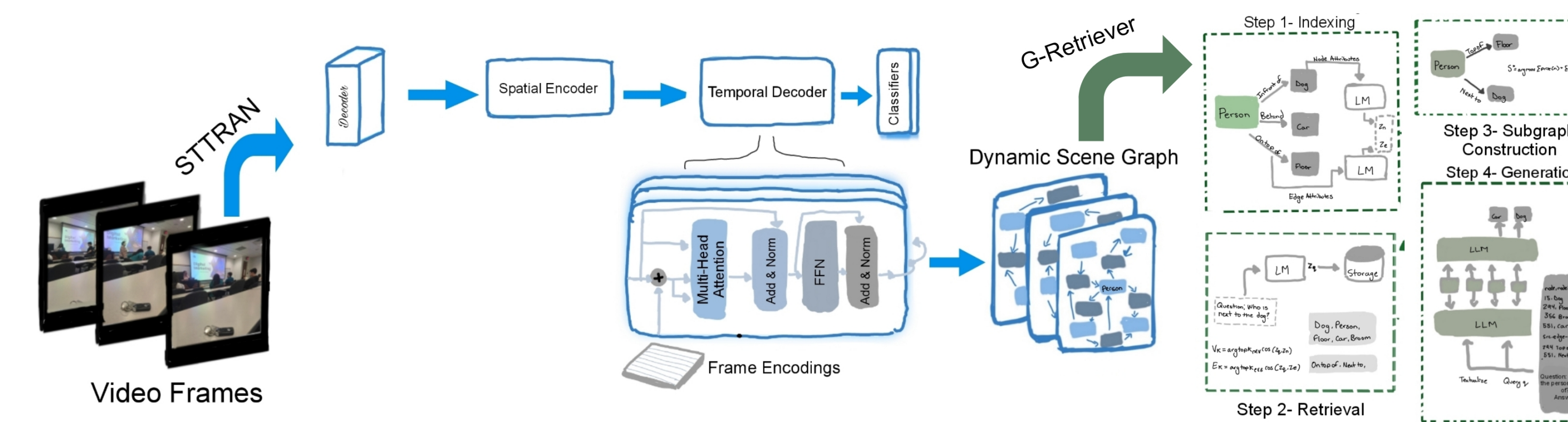


Figure 3. Full Pipeline

Results

Model Training: Our model was trained up to 10 epochs (early exit at 6) with a loss rate set at 1.0×10^{-5} and one warm up epoch (denoted as epoch 0), resulting in high loss values at the start.

Model Metrics: Trained on a comprehensive subset of 40,000 GQA samples, our model achieved a test accuracy of 81.25%, improving from an initial training loss of 4.6233 to 0.3626—a 92.16% reduction in loss. When evaluated on 20,000 held-out examples spanning 1,878 possible answers (vs. 48 in prior work), the model shows strong generalization across diverse object categories.



Figure 4. Loss Over Time

Analysis

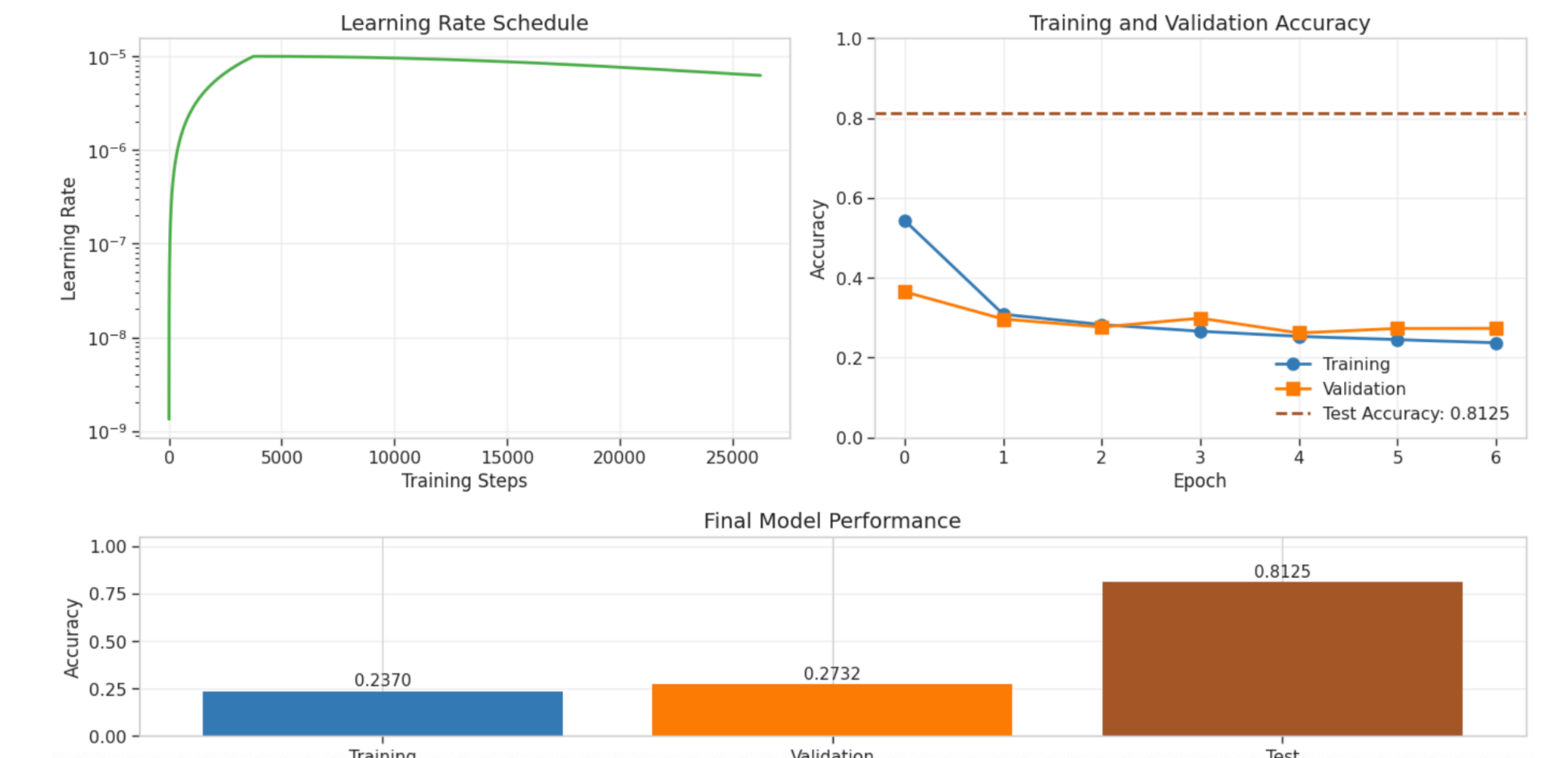


Figure 5. Final Model Performance

Loss Reduction: The training dynamics, i.e. the early stop at epoch 6, demonstrate that the model converges quickly and avoids overfitting when fine-tuned on the GQA dataset.

Key Insights: By leveraging a much larger label space, our approach achieves more object coverage and substantially higher intent-alignment rates compared to prior methods with limited categories. Our trade-off in compute time enables richer commonsense reasoning and deeper scene understanding across a wider range of scenarios.

Conclusion

We developed a live video querying framework that combines recent developments in transformer architecture and GNNs for application in blind and visually impaired accessibility efforts. Our method leverages a transformer architecture to generate scene graphs based on object relationships over time, which are then put into an optimized, queryable format for natural-language intent. We observed a significant reduction in loss during training, demonstrating strong generalization across diverse object categories.

References

- He, X., Tian, Y., Sun, Y., Chawla V., N., Laurent, T., LeCun, Y., Bresson, X., & Hooi, B. (2024, May). G-Retriever: Retrieval-Augmented Generation for Textual Understanding and Question Answering. arXiv.org. <https://arxiv.org/abs/2402.07630>
- Cong, Y., Wentong, L., Ackermann, H., Rosenhahn, B., Yang Y., M. (2021, August). Spatio-Temporal Transformer for Dynamic Scene Graph Generation. arXiv.org. <https://arxiv.org/abs/2107.12309>